

OE-EC704B — Optimization Technique

Semester VII • 3L:0T:0P • 3 Credits

Module 4 Exam Notes: Multivariable Optimization Algorithms

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1. Optimality Criteria for Multivariable Functions

Let $f(\mathbf{x})$ be a continuous multivariable function defined on the design vector $\mathbf{x} = [x_1, x_2, \dots, x_n]^T$.

1.1 Necessary Condition (First-Order)

Theorem: Multivariable Necessary Condition

If $f(\mathbf{x})$ is differentiable, a necessary condition for a point $\mathbf{x}^*$ to be a local extremum (minimum or maximum) is that the gradient vector evaluated at $\mathbf{x}^*$ must vanish:

$$\nabla f(\mathbf{x}^*) = \mathbf{0} \implies \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right]_{\mathbf{x}=\mathbf{x}^*}^T = \mathbf{0} \quad (1)$$

Points satisfying this are called **Stationary Points**.

1.2 Sufficient Condition (Second-Order)

Optimality is determined by evaluating the **Hessian Matrix** $H(\mathbf{x}^*)$, defined as the matrix of second-order partial derivatives:

$$H(\mathbf{x}^*) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right]_{n \times n} \quad (2)$$

Theorem: Multivariable Sufficient Condition

At the stationary point $\mathbf{x}^*$:

- If $H(\mathbf{x}^*)$ is **Positive Definite** (all eigenvalues > 0), then $\mathbf{x}^*$ is a **local minimum**.
- If $H(\mathbf{x}^*)$ is **Negative Definite** (all eigenvalues < 0), then $\mathbf{x}^*$ is a **local maximum**.
- If some eigenvalues are positive and others negative, $\mathbf{x}^*$ is a **Saddle Point**.

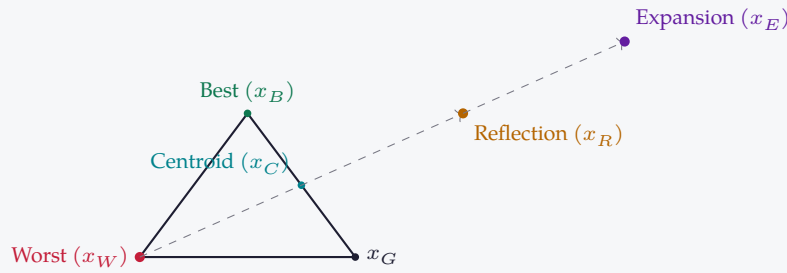
2. Direct Search Methods (No Gradients)

Direct search methods do not compute derivatives, making them suitable for noisy or discontinuous functions.

2.1 Simplex Search Method (Nelder-Mead)

A simplex in n -dimensional space is a geometric figure defined by $n + 1$ vertices (e.g., a triangle in 2D, a tetrahedron in 3D). The method moves the simplex toward the optimum using reflection, expansion, contraction, and shrinkage operations.

Nelder-Mead Simplex Operations in 2D



2.2 Hooke-Jeeves Pattern Search Method

Hooke-Jeeves is a conceptual search method consisting of two primary components:

1. **Exploratory Move:** Performs local searches along the coordinate axes to find a direction of descent.
2. **Pattern Move:** Performs an accelerated step along the direction established by connecting successive base points, bypassing slow orthogonal paths.

3. Gradient-Based Descent Methods

3.1 Cauchy's (Steepest Descent) Method

Steepest descent moves in the direction of the local negative gradient vector, which represents the direction of maximum rate of decrease of the function.

Steepest Descent Formulation

The iterative update equation is:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k) \quad (3)$$

Where:

- $\nabla f(\mathbf{x}_k)$ is the gradient vector at step k .
- α_k is the step length, determined by solving a unidirectional line search problem:

$$\text{Minimize } f(\mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k)) \quad \text{w.r.t. } \alpha \quad (4)$$

***Characteristics:** Simple to implement but exhibits slow convergence near the optimum due to orthogonal zig-zagging trajectories.

3.2 Newton's Multivariable Method

Newton's method uses a second-order quadratic approximation of the objective function.

Newton's Multivariable Update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - [H(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k) \quad (5)$$

Where $[H(\mathbf{x}_k)]^{-1}$ is the inverse of the Hessian matrix.

Characteristics: Converges quadratically (extremely fast near the optimum) but requires calculating and inverting the Hessian matrix at every step, which is computationally expensive for large systems.

4. Multi-Objective and Pareto Optimization

In real-world engineering, we often optimize multiple conflicting objectives simultaneously (e.g., minimizing cost while maximizing safety).

4.1 Pareto Dominance & Optimality

Instead of a single optimum, multi-objective optimization yields a set of trade-off solutions.

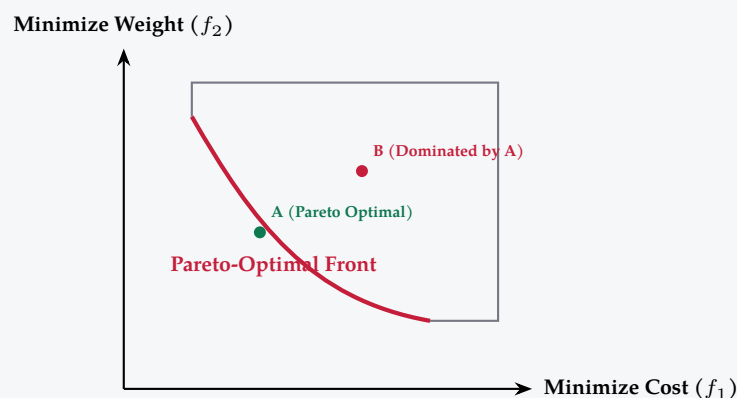
Definition of Pareto Dominance

A decision vector $\mathbf{x}_1$ is said to **dominate** $\mathbf{x}_2$ if:

1. $\mathbf{x}_1$ is no worse than $\mathbf{x}_2$ in all objectives.
2. $\mathbf{x}_1$ is strictly better than $\mathbf{x}_2$ in at least one objective.

A solution is **Pareto Optimal** if it is not dominated by any other feasible solution. The set of all Pareto optimal solutions in the objective space forms the **Pareto Front**.

Pareto Optimal Front (Conflicting Objectives)



Quick Revision — Module 4

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Key Concepts & Mathematical Rules

Gradient Vector (∇f)	$\nabla f = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right]^T = \mathbf{0}.$
Hessian Matrix (H)	$H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$ (square symmetric matrix).
Cauchy's Update	$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k)$ (Steepest Descent).
Newton's Update	$\mathbf{x}_{k+1} = \mathbf{x}_k - H^{-1} \nabla f(\mathbf{x}_k)$ (Quadratic Convergence).
Pareto Dominance Rule	$\mathbf{x}_1 \prec \mathbf{x}_2 \iff \forall i : f_i(\mathbf{x}_1) \leq f_i(\mathbf{x}_2) \text{ and } \exists j : f_j(\mathbf{x}_1) < f_j(\mathbf{x}_2).$

One-Line Definitions

- **Positive Definite Matrix:** A matrix whose eigenvalues are all strictly positive (indicates local minimum structure).
- **Saddle Point:** A stationary point at which the Hessian matrix is indefinite (eigenvalues of mixed signs).
- **Nelder-Mead Simplex:** A derivative-free search method using vertices of a geometric figure to crawl toward the optimum.
- **Hooke-Jeeves Method:** A coordinate-based search method combining local exploratory moves and pattern acceleration.
- **Steepest Descent:** An iterative gradient method moving in the direction of the local negative gradient vector.
- **Pareto Front:** The set of all non-dominated solutions representing optimal trade-offs in multi-objective problems.

Frequently Asked / Viva Questions

Q1. State the necessary and sufficient conditions for a local minimum of a multivariable function.

Necessary condition: The gradient vector must vanish, $\nabla f(\mathbf{x}^*) = \mathbf{0}$. Sufficient condition: The Hessian matrix $H(\mathbf{x}^*)$ must be positive definite.

Q2. What is a Hessian matrix and what is its role in multivariable optimization?

The Hessian is a square symmetric matrix of second-order partial derivatives. Its role is to provide local curvature information to identify if a stationary point is a minimum, maximum, or saddle point.

Q3. Explain why Cauchy's (Steepest Descent) method exhibits a zig-zag search trajectory.

Frequently Asked / Viva Questions (contd.)

Because successive search directions in Cauchy's method are mathematically orthogonal: $\mathbf{d}_k \cdot \mathbf{d}_{k-1} = 0$. This perpendicular stepping causes the algorithm to bounce back and forth in narrow valleys.

Q4. What are the operations performed in the Nelder-Mead simplex search method?

The operations are Reflection, Expansion, Contraction (Outer and Inner), and Shrinkage.

Q5. Differentiate between exploratory and pattern moves in the Hooke-Jeeves method.

An exploratory move searches along the coordinate axes to find a local descending direction. A pattern move accelerates the search by stepping along the vector connecting successive base points.

Q6. Compare multivariable Newton's method with the Steepest Descent method.

Newton's method converges quadratically (very fast) but requires calculating and inverting the Hessian matrix. Steepest descent only requires the first-order gradient, making it computationally simpler per step but much slower to converge.

Q7. What is Pareto optimality in multi-objective optimization?

A solution is Pareto optimal if no other solution exists that improves one objective without simultaneously degrading at least one other objective.

Q8. Define the concept of a saddle point.

A saddle point is a stationary point where the gradient is zero, but the Hessian is indefinite, meaning the function increases in some directions and decreases in others.

Q9. What is the purpose of a unidirectional line search in multivariable algorithms?

To find the optimal step length α_k that minimizes the function along the selected search direction $\mathbf{d}_k$: $\min_{\alpha} f(\mathbf{x}_k + \alpha \mathbf{d}_k)$.

Q10. What makes a matrix positive definite using Sylvester's Criterion?

A symmetric matrix is positive definite if and only if all its leading principal minors (determinants of top-left submatrices) are strictly positive.